Benedikt J. **Koch**

PhD Candidate in Statistics, Harvard University

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Summary

PhD candidate in Statistics at Harvard University with a focus on causal inference, counterfactual decision theory, and policy learning. Degrees from the University of Oxford and ETH Zurich. Industry research experience at Netflix in machine learning and inference. Full-time data science experience at QuantCo.

Education

Harvard University — *Cambridge, MA*
PhD in Statistics — Sep 2023 – Present
- Advisor: Prof. Kosuke Imai
- Research in causal inference, with emphasis on experimentation, policy learning, and counterfactual decision theory

University of Oxford — *Oxford, GBR*
MSc in Statistical Science - GPA: 82.2/100 — Oct 2022 – Sep 2023

ETH Zurich — *Zurich, CHE*
MSc in Applied Mathematics - GPA: 5.92/6.0 — Sep 2019 – Dec 2021

University of Vienna — *Vienna, AUT*
BSc in Mathematics - GPA: 1.3/1.0 — Oct 2016 – Jul 2019

Publications

Published Papers

- Ergodic robust maximisation of asymptotic growth with stochastic factor processes
  Finance and Stochastics (2026)
  D. Itkin, **B. Koch**, M. Larsson, and J. Teichmann

Working Papers

- Policy Learning with Weak Signals
  B. Koch, W. Chou, A. Bibaut, and N. Kallus, 2026
- An Axiomatic Foundation for Decisions with Counterfactual Utility
  B. Koch, K. Imai, and T. Strzalecki, 2026
- Statistical Decision Theory with Counterfactual Loss
  B. Koch and K. Imai, 2025

Work Experience

Netflix — *Los Gatos, CA*
Machine Learning & Inference Research Intern — Jun 2026 – Aug 2026
- Developed a framework for policy learning under weak signals, leading to the paper *Policy Learning with Weak Signals*
- Established minimax-rate optimality under smoothness assumptions on treatment effects
- Demonstrated significant engagement gains over benchmark methods in two large-scale Netflix experiments

Netflix — *Boston, MA*
Machine Learning & Inference Research Contractor — Sep 2025 – May 2026
- Developed a fast, debiased, low-rank CATE extrapolation framework to personalize promotions for new titles
- Built a semiparametric, nuclear-norm-regularized model to pool weak signals across high-dimensional model embeddings
- Designed scalable estimation routines and inference diagnostics for policy learning and targeting decisions

Netflix — *Los Gatos, CA*
Experimentation & Causal Inference Intern — Jun 2025 – Aug 2025
- Estimated promotion lift with IV designs using randomized assignments across experiments
- Built doubly robust targeting policies for personalized promotions, validated in a live A/B test
- Prototyped low-rank assortment methods for large-scale recommendation optimization

QuantCo *Berlin, DEU*
Data Scientist Sep 2021 – Sep 2022

- Designed an ML-based health-insurance pricing model serving over 800,000 policyholders
- Built actuarial decision tools supporting allocation of €120 million in funds
- Re-engineered the pricing pipeline, reducing runtime by 100x and improving reproducibility

HSBC *London, GBR*
Data Science & Engineering Intern Jun 2021 – Aug 2021

- Developed an improved credit risk-scoring model for lending decisions
- Designed a data aggregation method that increased usable training data by 12x
- Built scalable data transformations and analysis pipelines in Apache Spark

Ministry of Defense *Vienna, AUT*
Mathematical Analyst Feb 2019

- Developed a trajectory prediction algorithm for defense analytics applications

Aon Austria *Vienna, AUT*
Data Analyst Aug 2018 – Sep 2018

- Analyzed insurance data to support actuarial modeling and risk assessment

Austrian Armed Forces *AUT*
Prospective Officer Sep 2015 – Present

- Military service to become an officer of the Austrian Reserve Forces (EF)
- Trained and instructed groups of up to 120 conscripts

Teaching Experience

Harvard University *Cambridge, MA*
Teaching Fellow

- STAT 210: Probability I
- STAT 234: Sequential Decision Making
- STAT 286: Causal Inference with Applications

ETH Zurich *Zurich, CHE*
Student Assistant

- Probability Theory and Statistics